

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

Mini Project: Decentralized Attendance

Aim:

The aim of this project is to develop a Decentralized Attendance System using Blockchain Technology for securely recording, managing, and viewing student attendance. The system is designed to provide a transparent, tamper-resistant, and reliable method of maintaining attendance records by storing them on a blockchain network.

The project will use a Smart Contract developed in Solidity to manage attendance records. An authorized Teacher or Administrator will be able to mark a student as Present or Absent using the student's blockchain wallet address. Once the attendance is recorded on the blockchain, the information can be retrieved through the web application.

Requirements

1. Mark Present

The teacher can mark a particular student as present using the student's wallet address.

2. Mark Absent

The teacher can mark a particular student as absent using the student's wallet address.

3. Check Attendance

The system can retrieve and display the attendance record of an individual student.

4. Class Attendance

The system can retrieve and display the attendance records of all students stored in the system.

5. Teacher Authorization

Only the authorized teacher wallet should be allowed to perform attendance marking operations.

6. Blockchain Storage

Attendance records will be stored using a Solidity smart contract deployed on the Sepolia Test Network.

7. Web Interface

A simple and user-friendly web page will be provided for interacting with the attendance smart contract.

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

Technologies Used

Solidity – Used to develop the blockchain smart contract.

Remix IDE – Used to write, compile, and deploy the smart contract.

MetaMask – Used for blockchain wallet connection and transaction authorization.

Sepolia Testnet – Used for testing and deploying the smart contract.

HTML – Used to create the frontend structure.

CSS – Used to design the user interface.

JavaScript / Ethers.js – Used to connect the web application with the blockchain smart contract.

GitHub – Used to store and share the project source code.

Main Functions

The smart contract will provide the following important functions:

markPresent(address student) – Marks the specified student as present.

markAbsent(address student) – Marks the specified student as absent.

getAttendance(address student) – Returns the attendance record of a particular student.

getClassAttendance() – Returns the attendance records of all students.

Expected Outcome

The proposed system will demonstrate how Blockchain and Smart Contract technology can be applied to attendance management. It will provide a secure, transparent, and tamper-resistant attendance record, while allowing the teacher to conveniently manage attendance through a web-based interface.

The project will also demonstrate the practical integration of Solidity, MetaMask, Sepolia, and a web frontend in a blockchain-based application.

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

Steps:

1. Contract Code

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract DecentralizedAttendance {

    // The account that deploys the contract is the teacher/admin.
    address public owner;

    struct AttendanceRecord {
        address student;
        bool present;
        bool marked;
        uint256 timestamp;
    }

    // Student address => attendance record
    mapping(address => AttendanceRecord) private attendance;

    // Used so that we can return all attendance records.
    address[] private students;

    event AttendanceMarked(
        address indexed student,
        bool present,
        uint256 timestamp
    );

    modifier onlyOwner() {
        require(
            msg.sender == owner,
            "Only the owner can mark attendance"
        );
        _;
    }

    constructor() {
        owner = msg.sender;
    }
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

/**
 * @dev Mark a student as present.
 */
function markPresent(address student) external onlyOwner {
    require(student != address(0), "Invalid student address");

    // Add student to the list only the first time.
    if (!attendance[student].marked) {
        students.push(student);
    }

    attendance[student] = AttendanceRecord({
        student: student,
        present: true,
        marked: true,
        timestamp: block.timestamp
    });

    emit AttendanceMarked(
        student,
        true,
        block.timestamp
    );
}

/**
 * @dev Mark a student as absent.
 */
function markAbsent(address student) external onlyOwner {
    require(student != address(0), "Invalid student address");

    // Add student to the list only the first time.
    if (!attendance[student].marked) {
        students.push(student);
    }

    attendance[student] = AttendanceRecord({
        student: student,
        present: false,
        marked: true,
        timestamp: block.timestamp
    });

    emit AttendanceMarked(
        student,

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        false,
        block.timestamp
    );
}

/**
 * @dev Get the attendance record for one student.
 */
function getAttendance(address student)
    external
    view
    returns (
        address studentAddress,
        bool present,
        bool marked,
        uint256 timestamp
    )
{
    AttendanceRecord memory record = attendance[student];

    return (
        record.student,
        record.present,
        record.marked,
        record.timestamp
    );
}

/**
 * @dev Get all attendance records.
 */
function getClassAttendance()
    external
    view
    returns (AttendanceRecord[] memory)
{
    AttendanceRecord[] memory records =
        new AttendanceRecord[](students.length);

    for (uint256 i = 0; i < students.length; i++) {
        records[i] = attendance[students[i]];
    }

    return records;
}

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
/**
 * @dev Return number of students currently recorded.
 */
function getStudentCount() external view returns (uint256) {
    return students.length;
}
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

2. Front end code (Html)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Decentralized Attendance</title>
```

```
<script src="https://cdn.jsdelivr.net/npm/ethers@6.13.5/dist/ethers.umd.min.js"></script>
```

```
<style>
```

```
* {
```

```
    box-sizing: border-box;
```

```
}
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    background: #f4f7fb;
```

```
    margin: 0;
```

```
    padding: 30px;
```

```
}
```

```
.container {
```

```
    max-width: 900px;
```

```
    margin: auto;
```

```
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

h1 {

color: #222;

text-align: center;

}

.card {

background: white;

padding: 25px;

margin-top: 20px;

border-radius: 12px;

box-shadow: 0 3px 12px rgba(0, 0, 0, 0.08);

}

input {

width: 100%;

padding: 12px;

margin: 8px 0 15px;

border: 1px solid #ccc;

border-radius: 6px;

font-size: 15px;

}

button {

padding: 11px 18px;

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
margin: 5px;  
  
border: none;  
  
border-radius: 6px;  
  
cursor: pointer;  
  
font-size: 15px;  
  
}
```

```
button:hover {  
  
    opacity: 0.85;  
  
}
```

```
button:disabled {  
  
    opacity: 0.5;  
  
    cursor: not-allowed;  
  
}
```

```
.connect {  
  
    background: #2563eb;  
  
    color: white;  
  
}
```

```
.present {  
  
    background: #16a34a;  
  
    color: white;  
  
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
.absent {
    background: #dc2626;
    color: white;
}
```

```
.show {
    background: #7c3aed;
    color: white;
}
```

```
.status-box {
    margin-top: 15px;
    padding: 12px;
    border-radius: 6px;
    background: #eef2ff;
    word-break: break-word;
}
```

```
.student-record {
    border: 1px solid #ddd;
    padding: 15px;
    margin-top: 10px;
    border-radius: 8px;
    background: #fafafa;
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

    word-break: break-word;
}

```

```

.present-status {
    color: green;
    font-weight: bold;
}

```

```

.absent-status {
    color: red;
    font-weight: bold;
}

```

```

.not-marked {
    color: gray;
    font-weight: bold;
}

```

```

#account,
#contractAddress {
    word-break: break-all;
    color: #333;
}

```

```

.success {

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
background: #dcfce7;  
  
color: #166534;  
  
}
```

```
.error {  
  
background: #fee2e2;  
  
color: #991b1b;  
  
}
```

```
.warning {  
  
background: #fef3c7;  
  
color: #92400e;  
  
}
```

```
.info {  
  
background: #eef2ff;  
  
color: #3730a3;  
  
}
```

```
.wallet-info {  
  
line-height: 1.8;  
  
}
```

```
</style>
```

```
</head>
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

<body>

<div class="container">

<h1>Decentralized Attendance</h1>

<div class="card">

<h2>Wallet</h2>

<button class="connect" id="connectButton" onclick="connectWallet()">Connect
MetaMask</button>

<div class="wallet-info">

<p>Account: Not connected</p>

<p>Network: Not connected</p>

<p>Contract: </p>

<p>Contract Owner: Not
loaded</p>

</div>

</div>

<div class="card">

<h2>Mark Attendance</h2>

<label for="studentAddress">Student Ethereum Address</label>

<input type="text" id="studentAddress" placeholder="0x...">

<button class="present" id="presentButton" onclick="markPresent()">Mark
Present</button>

<button class="absent" id="absentButton" onclick="markAbsent()">Mark
Absent</button>

</div>

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

<div class="card">

    <h2>Student Attendance</h2>

    <button class="show" onclick="getStudentAttendance()">Show Attendance</button>

    <div id="studentResult"></div>

</div>

```

```

<div class="card">

    <h2>Class Attendance</h2>

    <button class="show" onclick="getClassAttendance()">Show Attendance</button>

    <div id="classResult"></div>

</div>

```

```

<div class="card">

    <h2>Status</h2>

    <div id="status" class="status-box info">Please connect MetaMask.</div>

</div>

</div>

```

```

<script>

const CONTRACT_ADDRESS = "0x0DE13B1eCcfF505b2b0308488ebeEEf8eCf12EE3";

const SEPOLIA_CHAIN_ID = "0xaa36a7";

const CONTRACT_ABI = [

    "function owner() view returns (address)",

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

"function markPresent(address student)",

"function markAbsent(address student)",

"function getAttendance(address student) view returns (address studentAddress, bool
present, bool marked, uint256 timestamp)",

"function getClassAttendance() view returns (tuple(address student, bool present, bool
marked, uint256 timestamp)[])",

"function getStudentCount() view returns (uint256)"

];

```

```
let provider = null;
```

```
let signer = null;
```

```
let contract = null;
```

```
let currentAccount = null;
```

```
document.getElementById("contractAddress").innerText = CONTRACT_ADDRESS;
```

```

async function connectWallet() {
  try {
    if (!window.ethereum) {
      setStatus("MetaMask is not installed. Please install MetaMask.", "error");
      return;
    }

    setStatus("Connecting to MetaMask...", "info");

    const accounts = await window.ethereum.request({

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        method: "eth_requestAccounts"

    });

    if (!accounts || accounts.length === 0) {

        setStatus("No MetaMask account was selected.", "error");

        return;

    }

    currentAccount = accounts[0];

    provider = new ethers.BrowserProvider(window.ethereum);

    const network = await provider.getNetwork();

    const chainId = "0x" + Number(network.chainId).toString(16);

    if (chainId !== SEPOLIA_CHAIN_ID) {

        document.getElementById("network").innerText = "Wrong Network";

        setStatus("Wrong network. Please switch MetaMask to Sepolia.", "warning");

        return;

    }

    signer = await provider.getSigner();

    contract = new ethers.Contract(

        CONTRACT_ADDRESS,

        CONTRACT_ABI,

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        signer

    );

    document.getElementById("account").innerText = currentAccount;
    document.getElementById("network").innerText = "Sepolia";

    const owner = await contract.owner();
    document.getElementById("contractOwner").innerText = owner;

    if (currentAccount.toLowerCase() === owner.toLowerCase()) {
        setStatus(
            "MetaMask connected. You are the contract owner and can mark attendance.",
            "success"
        );
    } else {
        setStatus(
            "MetaMask connected, but this account is NOT the contract owner. You can view
attendance but cannot mark it.",
            "warning"
        );
    }
} catch (error) {
    console.error("Connection error:", error);
    setStatus(
        "Connection failed: " + getErrorMessage(error),

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        "error"

    );

}

}

function ensureConnected() {

    if (!contract) {

        setStatus("Please connect MetaMask first.", "warning");

        return false;

    }

    return true;

}

async function ensureOwner() {

    if (!contract) {

        return false;

    }

    try {

        const owner = await contract.owner();

        if (currentAccount.toLowerCase() !== owner.toLowerCase()) {

            setStatus(

                "Only the contract owner can mark attendance.",

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        "error"

    );

    return false;

}

    return true;
} catch (error) {
    console.error(error);

    setStatus("Unable to verify contract owner.", "error");

    return false;
}
}

```

```

async function markPresent() {
    if (!ensureConnected()) {
        return;
    }

```

```

    if (!(await ensureOwner())) {
        return;
    }

```

```

const student = document

    .getElementById("studentAddress")

    .value

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
.trim());
```

```
if (!ethers.isAddress(student)) {
    setStatus("Please enter a valid Ethereum address.", "error");
    return;
}
```

```
if (student.toLowerCase() === ethers.ZeroAddress.toLowerCase()) {
    setStatus("Zero address is not allowed.", "error");
    return;
}
```

```
try {
    disableAttendanceButtons(true);

    setStatus(
        "Sending transaction to mark student PRESENT...",
        "info"
    );
```

```
const tx = await contract.markPresent(student);
```

```
setStatus(
    "Transaction submitted. Waiting for blockchain confirmation...",
    "info"
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

    );

    await tx.wait();

    setStatus(
        "Student marked PRESENT successfully.",
        "success"
    );

    await getStudentAttendance();
} catch (error) {
    console.error("Mark present error:", error);

    setStatus(
        "Failed to mark present: " + getErrorMessage(error),
        "error"
    );
} finally {
    disableAttendanceButtons(false);
}
}

async function markAbsent() {
    if (!ensureConnected()) {
        return;
    }
}

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```
}
```

```
if (!(await ensureOwner())) {
    return;
}
```

```
const student = document
    .getElementById("studentAddress")
    .value
    .trim();
```

```
if (!ethers.isAddress(student)) {
    setStatus("Please enter a valid Ethereum address.", "error");
    return;
}
```

```
if (student.toLowerCase() === ethers.ZeroAddress.toLowerCase()) {
    setStatus("Zero address is not allowed.", "error");
    return;
}
```

```
try {
    disableAttendanceButtons(true);

    setStatus(
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        "Sending transaction to mark student ABSENT...",
        "info"
    );

    const tx = await contract.markAbsent(student);

    setStatus(
        "Transaction submitted. Waiting for blockchain confirmation...",
        "info"
    );

    await tx.wait();

    setStatus(
        "Student marked ABSENT successfully.",
        "success"
    );

    await getStudentAttendance();
} catch (error) {
    console.error("Mark absent error:", error);

    setStatus(
        "Failed to mark absent: " + getErrorMessage(error),
        "error"
    );
}

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

    );
  } finally {
    disableAttendanceButtons(false);
  }
}

```

```

async function getStudentAttendance() {
  if (!ensureConnected()) {
    return;
  }

```

```

const student = document
  .getElementById("studentAddress")
  .value
  .trim();

```

```

if (!ethers.isAddress(student)) {
  setStatus("Please enter a valid Ethereum address.", "error");
  return;
}

```

```

try {
  setStatus(
    "Reading attendance from blockchain...",
    "info"
  );

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

);

const record = await contract.getAttendance(student);

const studentAddress = record[0];

const present = record[1];

const marked = record[2];

const timestamp = record[3];

let statusHTML;

if (!marked) {

statusHTML = 'NOT MARKED';

} else if (present) {

statusHTML = 'PRESENT';

} else {

statusHTML = 'ABSENT';

}

let date = "N/A";

if (marked) {

date = new Date(

Number(timestamp) * 1000

).toLocaleString();

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

}

```
document.getElementById("studentResult").innerHTML = `
```

```
  <div class="student-record">
```

```
    <p>
```

```
      <strong>Student:</strong><br>
```

```
      ${escapeHTML(studentAddress)}
```

```
    </p>
```

```
    <p>
```

```
      <strong>Status:</strong>
```

```
      ${statusHTML}
```

```
    </p>
```

```
    <p>
```

```
      <strong>Marked:</strong>
```

```
      ${marked ? "Yes" : "No"}
```

```
    </p>
```

```
    <p>
```

```
      <strong>Timestamp:</strong>
```

```
      ${escapeHTML(date)}
```

```
    </p>
```

```
  </div>
```

```
`;
```

```
setStatus(
```

```
  "Attendance retrieved successfully.",
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        "success"

    );

} catch (error) {

    console.error("Get attendance error:", error);

    setStatus(

        "Failed to retrieve attendance: " + getErrorMessage(error),

        "error"

    );

}

}

```

```

async function getClassAttendance() {

    if (!ensureConnected()) {

        return;

    }

}

```

```

try {

    setStatus(

        "Loading class attendance from blockchain...",

        "info"

    );

}

```

```

const records = await contract.getClassAttendance();

const container = document.getElementById("classResult");

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

if (records.length === 0) {

    container.innerHTML = "<p>No attendance records found.</p>";

    setStatus(

        "No attendance records found.",

        "info"

    );

    return;

}

let html = "";

records.forEach(function(record, index) {

    const student = record[0];

    const present = record[1];

    const marked = record[2];

    const timestamp = record[3];

    let statusHTML;

    if (!marked) {

        statusHTML = '<span class="not-marked">NOT MARKED</span>';

    } else if (present) {

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        statusHTML = '<span class="present-status">PRESENT</span>';
    } else {
        statusHTML = '<span class="absent-status">ABSENT</span>';
    }

```

```

let date = "N/A";

```

```

if (marked) {
    date = new Date(
        Number(timestamp) * 1000
    ).toLocaleString();
}

```

```

html += `
    <div class="student-record">
        <p><strong>Student #${index + 1}</strong></p>
        <p>
            <strong>Address:</strong><br>
            ${escapeHTML(student)}
        </p>
        <p>
            <strong>Status:</strong>
            ${statusHTML}
        </p>
        <p>

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

        <strong>Marked:</strong>

        ${marked ? "Yes" : "No"}

    </p>

    <p>

        <strong>Timestamp:</strong>

        ${escapeHTML(date)}

    </p>

</div>

`;

});

container.innerHTML = html;

setStatus(

    records.length + " attendance record(s) loaded successfully.",

    "success"

);

} catch (error) {

    console.error("Class attendance error:", error);

    setStatus(

        "Failed to load class attendance: " + getErrorMessage(error),

        "error"

    );

}

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

}

```
function disableAttendanceButtons(disabled) {
    document.getElementById("presentButton").disabled = disabled;
    document.getElementById("absentButton").disabled = disabled;
}
```

```
function setStatus(message, type) {
    const status = document.getElementById("status");
    status.innerText = message;
    status.className = "status-box " + type;
}
```

```
function getErrorMessage(error) {
    if (!error) {
        return "Unknown error";
    }

    if (error.code === 4001) {
        return "Transaction rejected by user.";
    }
}
```

```
if (error.reason) {
    return error.reason;
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

```

if (error.shortMessage) {
    return error.shortMessage;
}

```

```

if (
    error.info &&
    error.info.error &&
    error.info.error.message
){
    return error.info.error.message;
}

```

```

if (
    error.data &&
    typeof error.data === "string"
){
    return error.data;
}

```

```

if (error.message) {
    return error.message;
}

```

```

return "Transaction failed.";

```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

}

```
function escapeHTML(value) {
    const div = document.createElement("div");
    div.textContent = String(value);
    return div.innerHTML;
}
```

```
if (window.ethereum) {
    window.ethereum.on("accountsChanged", function(accounts) {
        if (accounts.length === 0) {
            currentAccount = null;
            provider = null;
            signer = null;
            contract = null;

            document.getElementById("account").innerText = "Not connected";
            document.getElementById("contractOwner").innerText = "Not loaded";

            setStatus(
                "MetaMask disconnected.",
                "warning"
            );
        } else {
            window.location.reload();
        }
    });
}
```

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

}

});

window.ethereum.on("chainChanged", function() {

 window.location.reload();

});

}

</script>

</body>

</html>

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

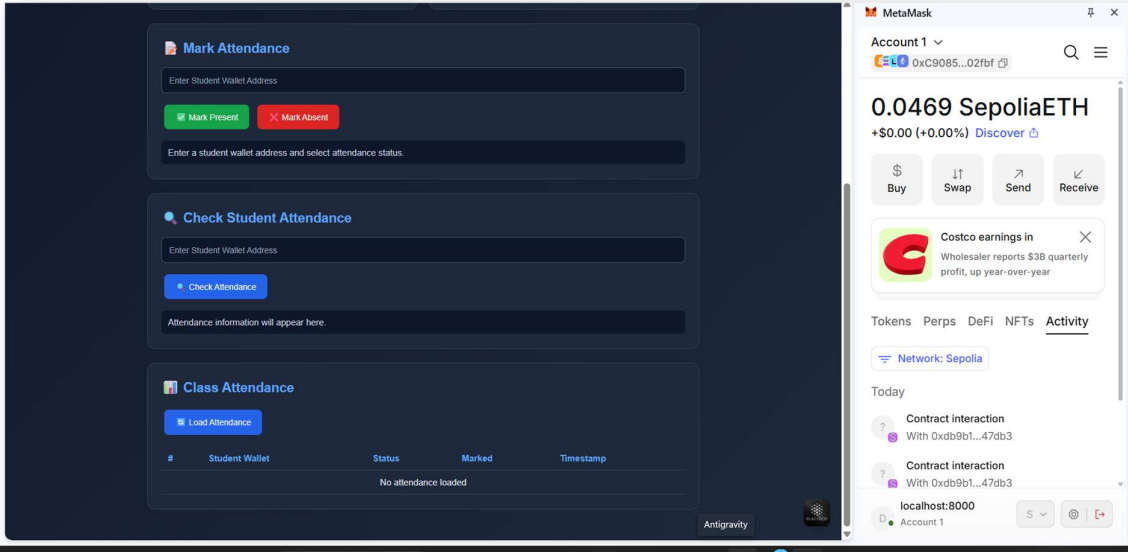
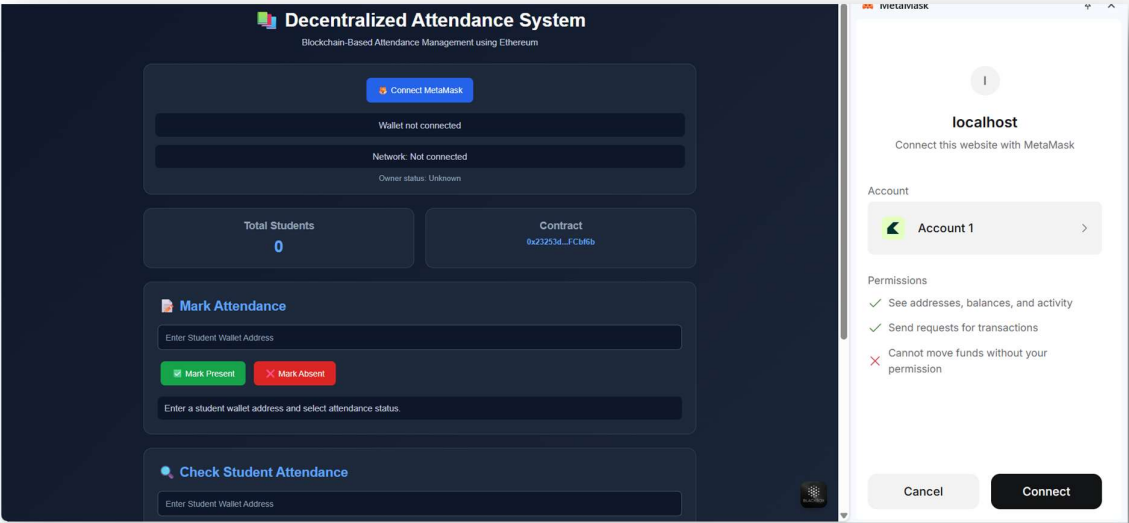
3. Test Cases:

TC	Test	Input	Expected Result
TC01	Connect Wallet	Click Connect MetaMask	MetaMask wallet connects successfully
TC02	Wrong Network	Connect wallet on a non-Sepolia network	System displays Wrong Network warning
TC03	Mark Present	Enter valid student address and click Mark Present	Student is marked PRESENT
TC04	Mark Absent	Enter valid student address and click Mark Absent	Student is marked ABSENT
TC05	Invalid Address	Enter invalid Ethereum address	Error message is displayed
TC06	Empty Address	Leave student address empty and click Mark Present	Error message is displayed
TC07	Student Attendance	Enter valid student address and click Show Attendance	Student attendance details are displayed
TC08	Class Attendance	Click Show Attendance under Class Attendance	All attendance records are displayed
TC09	Non-Owner Marking	Connect using non-owner wallet and mark attendance	System displays owner permission error
TC10	Transaction Confirmation	Submit attendance transaction	Transaction waits for blockchain confirmation
TC11	MetaMask Rejection	Reject transaction in MetaMask	Transaction rejected message is displayed
TC12	Wallet Disconnect	Disconnect MetaMask account	System displays wallet disconnected message
TC13	Network Change	Change MetaMask network	System reloads and checks the new network

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

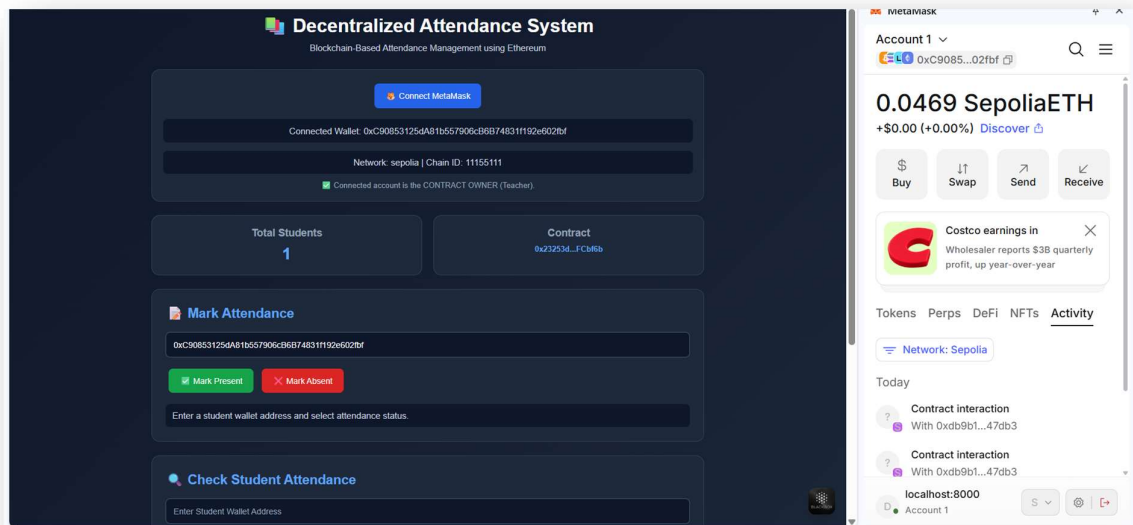
Output:

1. Front end screen



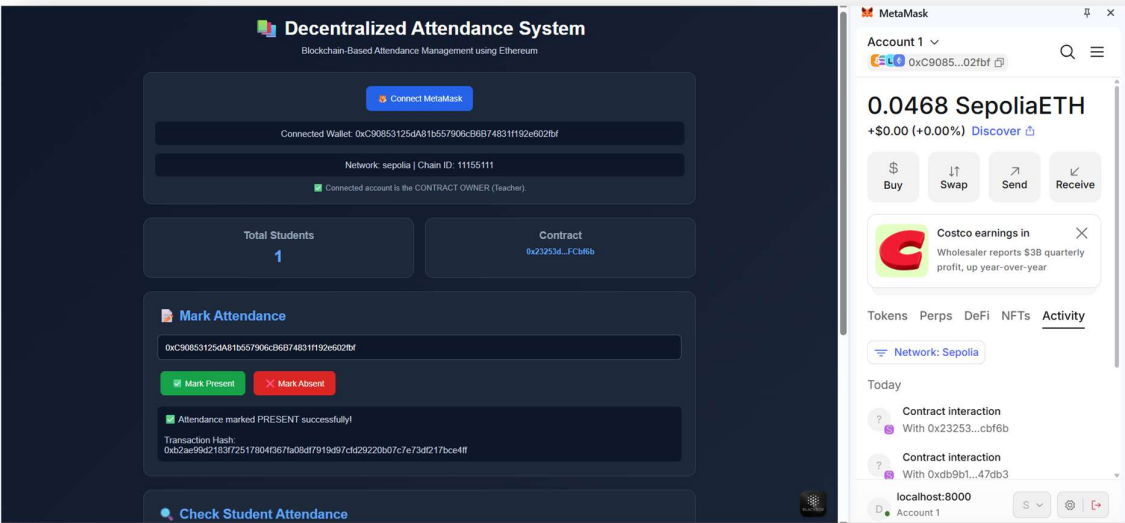
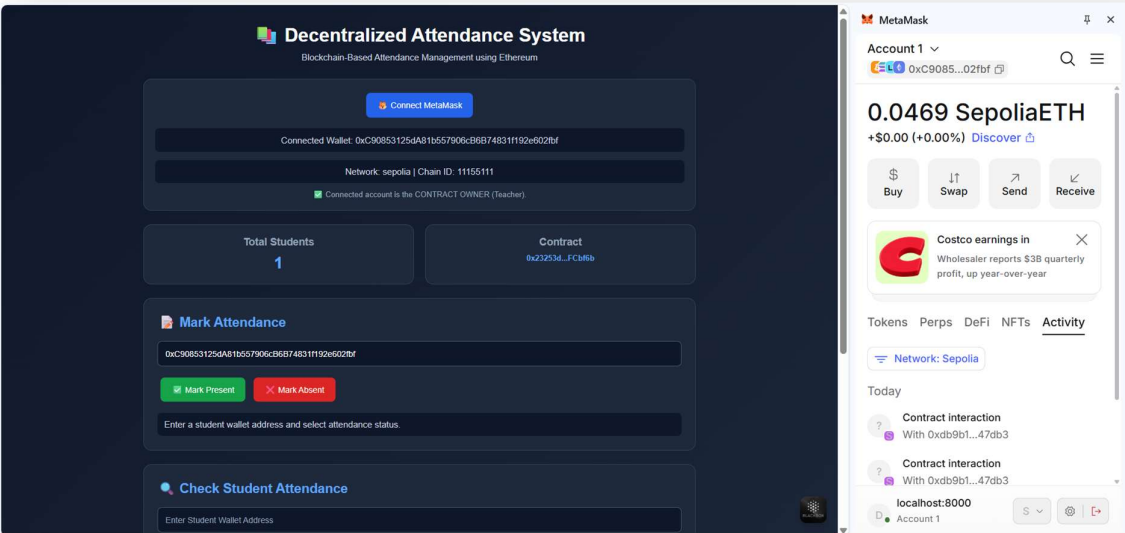
Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

2. Contract deployment



Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

3. Test Case: TC-03



Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

4. Test Case: TC-08

Decentralized Attendance System

Blockchain-Based Attendance Management using Ethereum

Connect MetaMask

Connected Wallet: 0xC90853125dA81b557906cB6B74831192e602bf

Network: sepolia | Chain ID: 11155111

Connected account is the CONTRACT OWNER (Teacher)

Total Students

1

Contract

0x23253d...fCb6b

Mark Attendance

0xC90853125dA81b557906cB6B74831192e602bf

Mark Present

Mark Absent

Attendance marked PRESENT successfully!

Transaction Hash: 0xb2ae99d2183f72517804f67fa08d7f919d97cd29220b07c7e73df217bce4ff

Check Student Attendance

MetaMask

Account 1

0xC9085...02fbf

0.0468 SepoliaETH

+ \$0.00 (+0.00%) Discover

Buy

Swap

Send

Receive

Costco earnings in

Wholesaler reports \$3B quarterly profit, up year-over-year

Tokens Perps DeFi NFTs Activity

Network: Sepolia

Today

Contract interaction

With 0x23253...cbf6b

Contract interaction

With 0xdb9b1...47db3

localhost:8000

Account 1

Mark Attendance

0xC90853125dA81b557906cB6B74831192e602bf

Mark Present

Mark Absent

Attendance marked PRESENT successfully!

Transaction Hash: 0xb2ae99d2183f72517804f67fa08d7f919d97cd29220b07c7e73df217bce4ff

Check Student Attendance

0xC90853125dA81b557906cB6B74831192e602bf

Check Attendance

Student: 0xC90853125dA81b557906cB6B74831192e602bf

Status: PRESENT

Marked Time: 9/25/2026, 5:25:36 PM

Class Attendance

Load Attendance

#	Student Wallet	Status	Marked	Timestamp
1	0xC90853125dA81b557906cB6B74831192e602bf	PRESENT	YES	9/25/2026, 5:25:36 PM

Dept: BE AIML	Year: 2026-27
Theem Engineering, Boisar	Subject: Blockchain technologies

5. Test Case TC-05:

